1️⃣ Principle of Working of a Transistor (Excess Carrier Injection)

A BJT works based on **minority carrier injection and control of charge carriers** across two p-n junctions.

Consider an **NPN transistor**:

- The **Emitter (N)** is heavily doped.
- The **Base (P)** is very thin and lightly doped.
- The **Collector (N)** is moderately doped.

Working Principle:

1. **Forward bias the Emitter-Base (EB) junction**
    - Majority carriers (electrons in NPN) are injected from the emitter into the base.
    - These injected electrons become **excess minority carriers** in the base region.
2. **Diffusion across Base**
    - Since the base is very thin and lightly doped:
        - Only a small number recombine with holes.
        - Most electrons diffuse across the base.
3. **Reverse bias the Collector-Base (CB) junction**
    - The electric field in the CB junction sweeps the electrons into the collector.
    - Thus, a large collector current flows.

Key Idea:

A **small base current controls a large collector current** because:

- The base only replaces the few recombined carriers.
- Most injected carriers reach the collector.

$$I_C \approx \beta I_B$$

where

- I_C= Collector current
- I_B= Base current
- β= Current gain

2️⃣ Modes of Operation of a Transistor (CE Configuration)

In **Common Emitter (CE)** configuration:

- Input: Base–Emitter
- Output: Collector–Emitter

The mode depends on biasing of the two junctions:

Mode	E-B Junction	C-B Junction	Application
Cutoff	Reverse	Reverse	OFF state
Active	Forward	Reverse	Amplifier
Saturation	Forward	Forward	Switch ON
Reverse Active	Reverse	Forward	Rare

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1. Cutoff Mode

- No base current.
- $I_C \approx 0$
- Transistor behaves like an open switch.

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2. Active Mode (Amplification Mode)

- EB: Forward biased
- CB: Reverse biased
- $I_C = \beta I_B$
- Used in amplifiers.

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3. Saturation Mode

- Both junctions forward biased.
- $V_{CE}$is very small (~0.2 V for silicon).
- Transistor behaves like a closed switch.

3️⃣ Constitution of Currents in a Transistor

In an **NPN transistor**, the currents are:

- I_E= Emitter current
- I_B= Base current
- I_C= Collector current

Current Relationship:

$$I_E = I_C + I_B$$

Current Components:

Emitter Current I_E

- Consists mainly of electrons injected into the base.
- Small component of holes injected from base to emitter.

Collector Current I_C

Has two parts:

1. **Main component:** Electrons injected from emitter and collected.
2. **Leakage current:** I_{CBO}(reverse leakage).

$$I_C = \alpha I_E + I_{CBO}$$

where $\alpha \approx 0.98 - 0.998$

Schematic Representation (NPN)

Emitter (N+) →→→ Base (P) →→→ Collector (N)

Electron flow: E → B → C
Base current: small (recombination current)

4️⃣ Ebers–Moll Model of a Transistor

The **Ebers–Moll model** is a large-signal model valid in all regions (cutoff, active, saturation).

It treats the transistor as:

- Two diodes
- Two controlled current sources

Equivalent Representation:

It consists of:

1. **Emitter-base diode**
2. **Collector-base diode**
3. **Controlled current source** $\alpha_F I_E$
4. **Controlled current source** $\alpha_R I_C$

Ebers-Moll Equations:

$$I_C = \alpha_F I_E - I_{CBO}$$
$$I_E = \alpha_R I_C - I_{EBO}$$

More complete form:

$$I_C = I_S\left(e^{V_{BE}/V_T} - 1\right) - \alpha_R I_S\left(e^{V_{BC}/V_T} - 1\right)$$

Where:

- I_S= Saturation current
- V_T= Thermal voltage
- α_F= Forward current gain
- α_R= Reverse current gain

Importance:

- Works in all operating regions.
- Used in transistor circuit simulation.

5️⃣ Small Signal Model of a Transistor

The **small signal model** is used for AC analysis in amplifier circuits.

Most commonly used model:
👉 **Hybrid-π Model**

Hybrid-π Model (CE Configuration)

Main elements:

1. r_π → Base-emitter resistance
2. $g_m v_\pi$ → Controlled current source
3. r_o → Output resistance

Important Parameters:

$$g_m = \frac{I_C}{V_T}$$
$$r_\pi = \frac{\beta}{g_m}$$
$$r_o = \frac{V_A}{I_C}$$

Where:

- g_m = Transconductance
- V_A = Early voltage

Small Signal Equivalent Circuit (CE)

```
    rπ
B ——/\/\/——┐
         |
         ↓ gm vπ
         |
C ————————┘— ro —— Ground
```

Why Small Signal Model?

- Used to calculate:
    - Voltage gain
    - Input impedance
    - Output impedance
- Valid for small AC variations around Q-point.